

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA5112

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)

Assessment Tool Weightage: 10%

QUESTIONS: 5 TOTAL MARKS: 25

Annexure A

Comparative Analysis

Code of Conduct:

I _E .Lakshmi Narayanan(192524249)_ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

1. SSL/TLS vs IPsec for Secure Communication

Scenario

A company has a web application accessed by employees through the Internet. The company needs to protect both **web traffic** and **network-layer communication**.

Applying SSL/TLS

SSL/TLS is implemented between the application and transport layers. For example:

Client Browser → TLS → Web Server

TLS provides:

- Confidentiality through encryption.
- Integrity using authentication mechanisms.
- Server authentication using digital certificates.
- Protection for HTTPS communication.

Applying IPsec

IPsec works at the **Network Layer (Layer 3)**.

Client Network → IPsec VPN → Company Network

IPsec can provide:

- Encryption of IP packets.
- Authentication.
- Integrity protection.
- Secure VPN communication.

Comparison

Feature	SSL/TLS	IPsec
Main layer	Transport/Application boundary	Network Layer
Main use	HTTPS, web applications	VPN and network communication
Protection	Application traffic	IP packets
Authentication	Digital certificates	Certificates/keys
Deployment	Application-specific	Network-wide
Best suited for	Web security	VPN/network security

Conclusion

For **web-based protection**, SSL/TLS is better because HTTPS directly protects browser-to-server communication.

For **network-layer protection**, IPsec is better because it can secure IP traffic between networks or hosts.

Therefore: TLS → Web/Application security; IPsec → Network/VPN security.

2. Secure Email System: End-to-End vs Gateway-Based Encryption

Scenario

An enterprise wants to protect confidential emails exchanged between employees and external customers.

Approach 1: End-to-End Encryption

In end-to-end encryption, the message is encrypted by the sender and decrypted only by the intended receiver.

Sender → Encrypted Email → Internet → Receiver

Examples include technologies such as S/MIME and PGP.

Advantages:

- Strong confidentiality.
- Intermediary servers cannot read the message.
- Protects data throughout transmission.
- Suitable for highly confidential information.

Disadvantages:

- Key management can be difficult.
- Users may need certificate/key configuration.
- Recovery and administration can be complicated.

Approach 2: Gateway-Based Encryption

A security gateway automatically encrypts and decrypts emails.

Sender → Enterprise Gateway → Encrypted Email → Receiver Gateway → Receiver

Advantages:

- Centralized management.
- Easier for administrators.
- Users do not need to manually encrypt every email.
- Easier policy enforcement.
- Suitable for large organizations.

Disadvantages:

- Gateway becomes a trusted point.
- If the gateway is compromised, email confidentiality can be affected.
- Does not provide the same strict endpoint-only protection as end-to-end encryption.

Comparison

Feature	End-to-End	Gateway-Based
Confidentiality	Very high	High
Administration	Difficult	Easy
Key management	User/device focused	Centralized
User involvement	Usually higher	Usually lower
Enterprise scalability	More complex	Excellent
Control	Endpoint-based	Centralized

Conclusion

For a large enterprise, **gateway-based encryption is generally more practical** because it provides centralized management, automatic policy enforcement, and easier deployment.

For highly sensitive communications, **end-to-end encryption provides stronger protection** because only the endpoints can decrypt the content.

3. HMAC vs Digital Signature

Scenario

A bank sends transaction information from its server to another trusted server.

Example:

Bank Server → Transaction Data → Receiving Server

The organization needs to ensure that the message has not been modified and comes from an authorized sender.

HMAC

HMAC uses a **shared secret key**.

$\text{HMAC} = \text{Hash}(\text{Message} + \text{Secret Key})$

The sender generates an HMAC and sends it with the message.

The receiver uses the same secret key to calculate the HMAC and compares the results.

Advantages:

- Fast.
- Efficient for large amounts of data.
- Provides integrity.
- Provides authentication between parties sharing the secret key.

Disadvantage:

- Both parties must securely share and protect the secret key.
- Does not provide non-repudiation.

Digital Signature

A digital signature uses **private and public keys**.

Sender:

$\text{Hash}(\text{Message}) \rightarrow \text{Sign with Private Key}$

Receiver:

Verify with Public Key

Advantages:

- Provides authentication.
- Provides integrity.
- Provides non-repudiation.

- No shared secret is required between sender and receiver.

Disadvantages:

- Slower than HMAC.
- Requires public-key infrastructure and key/certificate management.

Comparison

Feature	HMAC	Digital Signature
Key type	Shared secret	Private/Public key
Speed	Fast	Slower
Integrity	Yes	Yes
Authentication	Yes	Yes
Non-repudiation	No	Yes
Key management	Shared-key management	PKI/certificate management

Conclusion

HMAC is better when high-speed authentication and integrity are required between trusted systems.

Digital signatures are better when authentication, integrity, and **non-repudiation** are required.

Therefore, for legal or financial transactions, **digital signatures provide stronger authentication**, while HMAC is more efficient for high-volume internal communication.

4. Spam Detection: Supervised vs Unsupervised Learning

Scenario

An email provider wants to automatically classify incoming emails as:

- **Spam**
- **Not Spam (Ham)**

Supervised Learning

Supervised learning uses a dataset containing already-labelled emails.

Example:

Email	Label
"Win a free prize!"	Spam
"Meeting at 10 AM"	Ham
"Claim your reward now!"	Spam
"Project report attached"	Ham

The model learns patterns from these examples.

Common algorithms:

- Naive Bayes
- Decision Trees
- Support Vector Machines
- Neural Networks

Advantages

- Usually high classification accuracy.
- Can learn specific spam patterns.

- Easy to evaluate using labelled test data.

Disadvantages

- Requires a large labelled dataset.
- New spam patterns may require retraining.

Unsupervised Learning

Unsupervised learning works with **unlabelled emails**.

The system identifies groups or patterns automatically.

For example:

Email Dataset → Feature Extraction → Clustering → Spam-like Group / Normal Group

Techniques include:

- K-Means clustering.
- DBSCAN.
- Anomaly detection.

Advantages

- Does not require manually labelled data.
- Can discover previously unknown patterns.
- Useful for detecting new types of spam.

Disadvantages

- Classification accuracy can be lower.
- Determining which cluster represents spam can be difficult.
- More difficult to evaluate.

Comparison

Feature	Supervised	Unsupervised
Labelled data	Required	Not required
Accuracy	Generally higher	Generally lower
Detect known spam	Excellent	Good
Detect new patterns	Moderate	Good
Training data	Requires preparation	Easier to collect
Enterprise email filtering	Highly suitable	Useful as supplementary detection

Conclusion

Supervised learning is generally more effective for real-world spam filtering because large labelled email datasets are available and high classification accuracy is important.

A practical system can combine both approaches:

Supervised Classifier + Unsupervised Anomaly Detection → Improved Spam Detection

5. AES vs RSA for Secure Communication

Scenario

A company needs to securely transmit a large confidential file between two systems.

A hybrid encryption framework can use **AES for data encryption** and **RSA for secure key exchange**.

AES

AES is a **symmetric encryption algorithm**.

The same secret key is used for encryption and decryption.

Plaintext+AES Key→Ciphertext Ciphertext+AES Key→Plaintext

Advantages:

- Very fast.
- Efficient for large files.
- Low computational overhead.
- Suitable for bulk data encryption.

RSA

RSA is an **asymmetric encryption algorithm**.

It uses:

- Public key.
- Private key.

RSA can be used to securely exchange or protect a symmetric session key.

Example:

AES Session Key RSA Public Key Encrypted Key

The receiver uses the RSA private key to recover the AES key.

Comparison

Feature	AES	RSA
Type	Symmetric	Asymmetric
Keys	One shared key	Public + private
Speed	Very fast	Slower
Large data encryption	Excellent	Not suitable
Key exchange	Requires secure key distribution	Useful for key establishment
Computational cost	Low	High

Secure Communication Framework

A practical system works as follows:

Step 1: Generate a random AES session key.

Step 2: Encrypt the actual message/file using AES.

Step 3: Encrypt/protect the AES session key using the receiver's RSA public key.

Step 4: Send:

Encrypted Data + Encrypted AES Key

Step 5: Receiver uses the RSA private key to recover the AES session key.

Step 6: Receiver uses AES to decrypt the original data.

Conclusion

AES provides **efficient confidentiality for large amounts of data**, while RSA provides secure **key establishment** and public-key functionality.

Therefore, the best solution is a **hybrid AES + RSA system**:

RSA for Key Exchange + AES for Data Encryption

This combines the **security advantages of asymmetric cryptography with the speed of symmetric encryption**.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand